

CS4780 - Lab 4: One- and Two-way ANOVA in R

Contents

Introduction to Analysis of Variance	2
The one-way ANOVA	2
Assumptions of ANOVA test	2
Computing a one-way ANOVA	2
Steps in a one-way ANOVA	4
One-way ANOVA in R	4
Checking the data	4
Visualise the data	5
Exercise 1	6
Performing a one-way ANOVA test	6
Interpret the result of one-way ANOVA tests	6
Multiple pairwise-comparison between the means of groups	7
Exercise 2	7
Checking ANOVA assumptions	8
Non-parametric alternative to one-way ANOVA test	9
The two-way ANOVA	10
Two-way ANOVA test hypotheses	10
Assumptions of two-way ANOVA test	10
Compute two-way between-subject ANOVA R	10
Compute summary statistics	11
Visualise the data	11
Computing the two-way ANOVA	13
Multiple pairwise-comparison between the means of groups	14
Checking ANOVA assumptions	14
Exercise 3	15
Further reading	15

Submitting your Lab assignment

Add the answers for the individual exercises to `Lab04_Worksheet.txt`. At the end of the lab, please submit the filled worksheet as **plain text** via the corresponding link on Blackboard. *No other formats will be considered.*

After submission deadline, the answers to the exercises will be published as `Lab04_Solutions.txt` on Blackboard.

Note This tutorial will make use of the following packages:

- `dplyr`
- `gplots`
- `ggplot2`
- `car`

If your R installation does not have these packages installed, please install them with the `install.packages()` command, e.g. `install.packages("dplyr")`.

Introduction to Analysis of Variance

Analysis of Variance (ANOVA) helps you test differences between two or more group means. ANOVA test is centred around the different sources of variation (variation between and within group) in a typical variable. A primary ANOVA test provides evidence of the existence of the mean equality between the group. This statistical method is an extension of the t-test. It is used in a situation where the factor variable has more than one group.

The one-way ANOVA

The **one-way analysis of variance (ANOVA)**, also known as *one-factor ANOVA*, is an extension of **independent two-samples t-test** for comparing means in a situation where there are more than two groups. In **one-way ANOVA**, the data is organized into several groups based on one single grouping variable (also called *factor* variable). This tutorial describes the basic principle of the **one-way ANOVA test** and provides practical **anova test examples** in R software.

ANOVA test hypotheses:

- Null hypothesis: the means of the different groups are the same
- Alternative hypothesis: At least one sample mean is not equal to the others.

Note that, if you have only two groups, you can use **t-test**. In this case, the F-test and the t-test are equivalent.

Assumptions of ANOVA test

Here we describe the requirement for **ANOVA test**. ANOVA test can be applied only when:

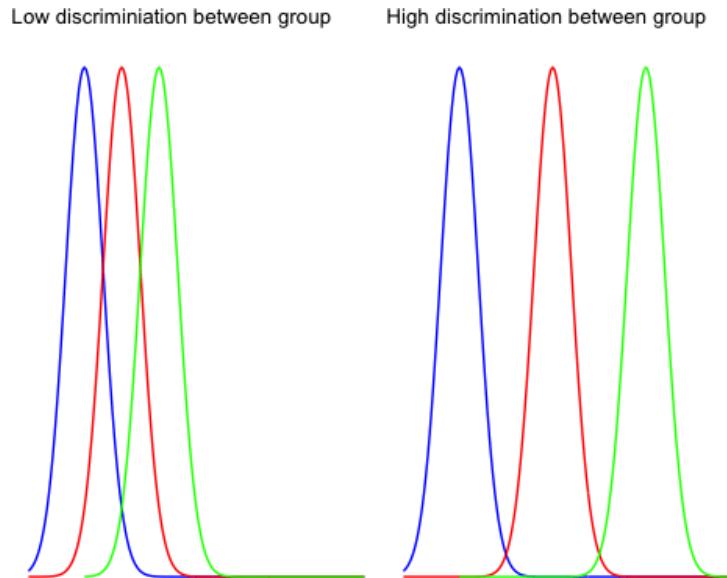
- The observations are obtained independently and randomly from the population defined by the factor levels
- The data of each factor level are normally distributed.
- These normal populations have a common variance. (**Levene's test** can be used to check this.)

Computing a one-way ANOVA

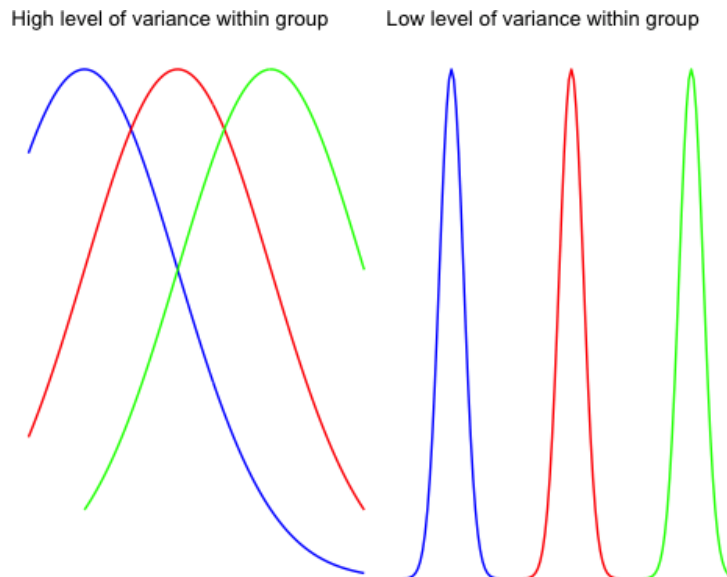
The F-statistic is used to test if the data are from significantly different populations, i.e., different sample means. To compute the F-statistic, you need to divide the **between-sample variance** over the **within-sample variance**. The **between-sample** variance reflects the differences between the samples inside all of the population. Look at the two graphs below to understand the concept of between-sample variance. The

left graph shows very little variation between the three samples, and it is very likely that the three means tends to the **overall** mean (i.e., mean for the three samples).

The right graph plots three distributions far apart, and none of them overlap. There is a high chance the difference between the total mean and the samples mean will be large.



The **within sample** variance considers the difference between the samples. The variation comes from the individual observations; some points might be totally different than the sample means. The **within sample** variance picks up this effect and refer to the sampling error. To understand visually the concept of within sample variance, look at the graph below. The left part plots the distribution of three different samples. You increased the spread of each sample and it is clear the individual variance is large. The F-test will decrease, meaning you tend to accept the null hypothesis. The right part shows exactly the same samples (identical mean) but with lower variance. It leads to an increase of the F-test and tends in favour of the alternative hypothesis.



You can use both measures to construct the F-statistics. It is very intuitive to understand the F-statistic. If

the numerator increases, it means the between-sample variance is high, and it is likely the samples in the sample are drawn from completely different distributions.

In other words, a low F-statistic indicates little or no significant difference between the sample's average.

Steps in a one-way ANOVA

Assume that we have 3 groups (A, B, C) to compare:

1. Compute the common variance, which is called **variance within samples** (S^2_{within}) or **residual variance**.
2. Compute the variance between sample means as follow:
 - Compute the mean of each group
 - Compute the **variance between sample means** ($S^2_{between}$)
3. Produce F-statistic as the ratio of ($\frac{S^2_{within}}{S^2_{between}}$).

Note that, a lower ratio (ratio < 1) indicates that there are no significant difference between the means of the samples being compared. However, a higher ratio implies that the variation among group means are significant.

One-way ANOVA in R

Here, we'll use the built-in R data set named *PlantGrowth*. It contains the weight of plants obtained under a control and two different treatment conditions.

```
my_data <- PlantGrowth
```

Checking the data

Use the function `summary([...])` to get an idea of what the data looks like.

In R terminology, the column 'group' is called factor and the different categories ('ctr', 'trt1', 'trt2') are named factor levels. **The levels are ordered alphabetically.**

```
# Show the levels
levels(my_data$group)
```

```
## [1] "ctrl" "trt1" "trt2"
```

If the levels are not automatically in the correct order, re-order them as follow:

```
library(dplyr)
my_data <- my_data %>% mutate(group=factor(group, order=TRUE))
```

Let's now compute the descriptive statistics (mean and sd) by groups using the 'dplyr' package:

```
library(dplyr)
group_by(my_data, group) %>%
  summarise(
    count = n(),
    mean = mean(weight, na.rm = TRUE),
    sd = sd(weight, na.rm = TRUE)
  )
```

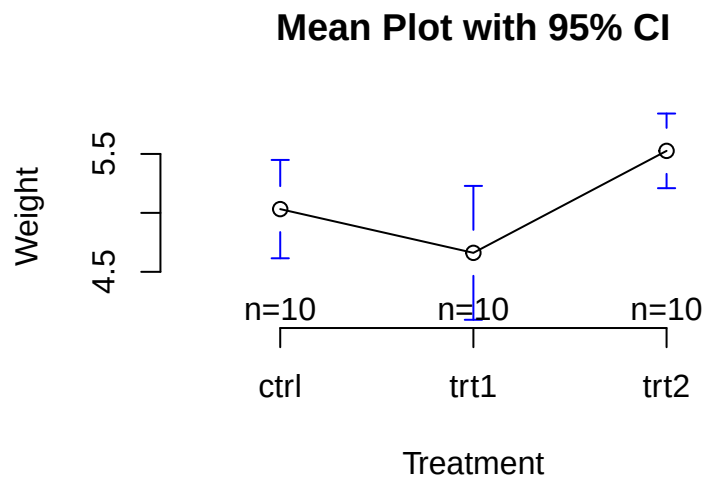
```
## # A tibble: 3 x 4
##   group count mean   sd
##   <ord> <int> <dbl> <dbl>
## 1 ctrl     10  5.03 0.583
```

```
## 2 trt1      10  4.66 0.794
## 3 trt2      10  5.53 0.443
```

Visualise the data

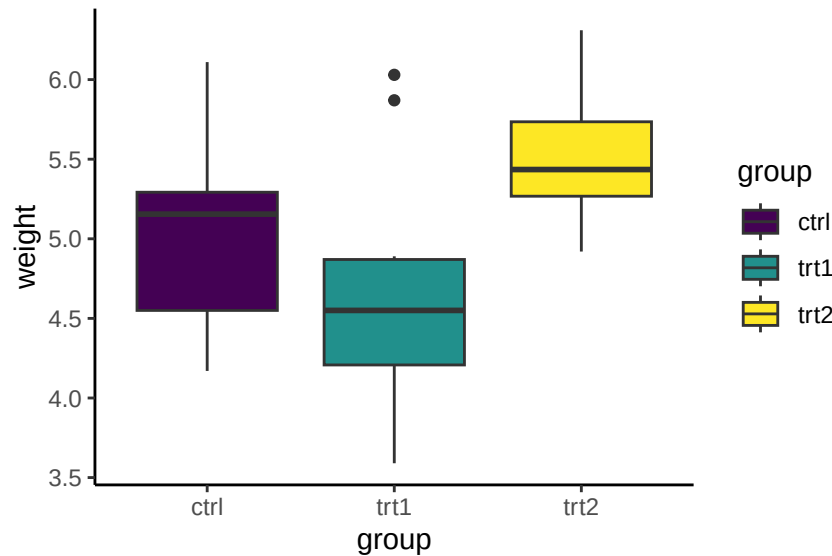
It is always a good idea to start an analysis by looking at the data. For this we can for example plot the means and the 95% confidence interval:

```
# plotmeans
library("ggplots")
plotmeans(weight ~ group, data = my_data, frame = FALSE,
          xlab = "Treatment", ylab = "Weight",
          main="Mean Plot with 95% CI")
```



Another popular way to visualise data for a one-way ANOVA is to use boxplot. The 'ggplot2' package provides a flexible way to render boxplots and control the graphics:

```
# ADAPT
library(ggplot2)
ggplot(my_data, aes(x = group, y = weight, fill = group)) +
  geom_boxplot() +
  theme_classic()
```



Exercise 1

For this exercise we will use the `iris` dataset. For details check `?iris`. Complete the following tasks:

1. Assign the contents of “iris” to “my_data”
2. Print the number of levels of the factor “Species”
3. Using ggplot, create a boxplot of “Species” vs “Sepal.Width”

Performing a one-way ANOVA test

We want to know if there is any significant difference between the average weights of plants in the 3 experimental conditions.

The R function `aov()` can be used to answer to this question. The function `summary.aov()` is used to summarize the analysis of variance model.

```
# Compute the analysis of variance
res.aov <- aov(weight ~ group, data = my_data)
# Summary of the analysis
summary(res.aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## group         2  3.766   1.8832   4.846 0.0159 *
## Residuals    27 10.492   0.3886
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The output includes the columns *F value* and *Pr(>F)* corresponding to the p-value of the test.

Interpret the result of one-way ANOVA tests

As the p-value is less than the significance level 0.05, we can conclude that there are significant differences between the groups highlighted with “*” in the model summary.

Multiple pairwise-comparison between the means of groups

In one-way ANOVA test, a significant p-value indicates that some of the group means are different, but we don't know which pairs of groups are different. It is possible to perform multiple pairwise-comparison, to determine if the mean difference between specific pairs of group are statistically significant.

Tukey multiple pairwise-comparisons

As the ANOVA test is significant, we can compute **Tukey HSD** (Tukey Honest Significant Differences, R function: `TukeyHSD()`) for performing multiple pairwise-comparison between the means of groups.

The function `TukeyHSD()` takes the fitted ANOVA as an argument.

```
TukeyHSD(res.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = weight ~ group, data = my_data)
##
## $group
##          diff          lwr          upr      p adj
## trt1-ctrl -0.371 -1.0622161 0.3202161 0.3908711
## trt2-ctrl 0.494 -0.1972161 1.1852161 0.1979960
## trt2-trt1 0.865 0.1737839 1.5562161 0.0120064
```

- **diff**: difference between means of the two groups
- **lwr**, **upr**: the lower and the upper end point of the confidence interval at 95% (default)
- **p adj**: p-value after adjustment for the multiple comparisons.

It can be seen from the output, that only the difference between trt2 and trt1 is significant with an adjusted p-value of 0.012.

Pairwise t-test

The function `pairwise.t.test(...)` can be also used to calculate pairwise comparisons between group levels with corrections for multiple testing.

```
pairwise.t.test(my_data$weight, my_data$group,
                p.adjust.method = "BH")
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data: my_data$weight and my_data$group
##
##      ctrl  trt1
## trt1 0.194 -
## trt2 0.132 0.013
##
## P value adjustment method: BH
```

The result is a table of p-values for the pairwise comparisons. Here, the p-values have been adjusted by the Benjamini-Hochberg method.

Exercise 2

Usig the “iris” dataset, complete the following tasks:

1. Compute an ANOVA for the factor “Species” and the dependent variable “Sepal.Width”
2. Describe if there is a significant effect or not.
3. Run a pair-wise t-test for the above ANOVA. **But** use a Bonferroni correction instead of the Benjamini-Hochberg method.
4. Describe which pairings show significant differences

Checking ANOVA assumptions

The ANOVA test assumes that the data are normally distributed and the variance across groups are homogeneous. We can check that with some diagnostic plots.

Homogeneity of variance

A number of methods can be used to check for homogeneity of variances, e.g. the *Bartlett's* or *Levene's* test. Here we will use the *Levene's test* that is less sensitive to departures from normal distribution. Using the function `leveneTest()` (“car” package):

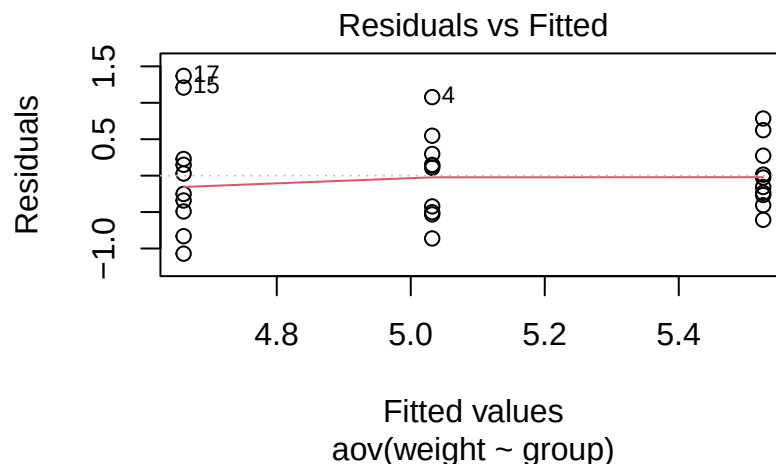
```
library(car)
leveneTest(weight ~ group, data = my_data)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  2  1.1192 0.3412
##      27
```

From the output above we can see that the p-value is not less than the significance level of 0.05. This means that there is no evidence to suggest that the variance across groups is statistically significantly different. Therefore, we can assume the homogeneity of variances in the different treatment groups.

The **residuals versus fits plot** can be used to check the homogeneity of variances. In the plot below, there is no evident relationships between residuals and fitted values (the mean of each groups), which is good. So, we can assume the homogeneity of variances.

```
# 1. Homogeneity of variances
plot(res.aov, 1)
```



Note that points 17, 15, 4 are detected as outliers, which can severely affect normality and homogeneity of variance. It can be useful to remove outliers to meet the test assumptions.

Check the normality assumption

As for the homogeneity of variances, a number of tests exist to check for normality such as the Kolmogorov–Smirnov test. Here we will use the *Shapiro-Wilk test*:

```
# Extract the residuals
aov_residuals <- residuals(object = res.aov )
# Run Shapiro-Wilk test
shapiro.test(x = aov_residuals )
```

```
##
##  Shapiro-Wilk normality test
##
## data:  aov_residuals
## W = 0.96607, p-value = 0.4379
```

The test results of $W = 0.96$, $p = 0.6$ mean that there is no indication that normality is violated.

Non-parametric alternative to one-way ANOVA test

Note that, a non-parametric alternative to one-way ANOVA is **Kruskal-Wallis rank sum test**, which can be used when ANNOVA assumptions are not met.

```
kruskal.test(weight ~ group, data = my_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  weight by group
## Kruskal-Wallis chi-squared = 7.9882, df = 2, p-value = 0.01842
```

The two-way ANOVA

Two-way ANOVA test is used to evaluate simultaneously the effect of two grouping variables (A and B) on a response variable.

The grouping variables are also known as factors. The different categories (groups) of a factor are called levels. The number of levels can vary between factors. The level combinations of factors are called cell.

- When the sample sizes within cells are equal, we have the so-called **balanced design**. In this case, the standard two-way ANOVA test can be applied.
- When the sample sizes within each level of the independent variables are not the same (case of **unbalanced designs**), the ANOVA test should be handled differently.

Two-way ANOVA test hypotheses

1. There is no difference in the means of factor A
2. There is no difference in means of factor B
3. There is no interaction between factors A and B

The alternative hypothesis for cases 1 and 2 is: the means are not equal.

The alternative hypothesis for case 3 is: there is an interaction between A and B.

Assumptions of two-way ANOVA test

Two-way ANOVA, like all ANOVA tests, assumes that the observations within each cell are normally distributed and have equal variances. We'll show you how to check these assumptions after fitting ANOVA.

Compute two-way between-subject ANOVA R

We will use the built-in R data set named *ToothGrowth*. It contains data from a study evaluating the effect of vitamin C on tooth growth in Guinea pigs. The experiment has been performed on 60 pigs, where each animal received one of three dose levels of vitamin C (0.5, 1, and 2 mg/day) by one of two delivery methods, (orange juice or ascorbic acid (a form of vitamin C and coded as VC). Tooth length was measured and a sample of the data is shown below. In our example we have a balanced design, meaning that there is an equal sample sizes within levels of our independent groupings. And because each animal only received one of the treatments, we have a between-subject design.

What we first want to determine is if tooth length depends on the two factors of supply and dose.

Let's first get our data and look at the structure:

```
my_data <- ToothGrowth
str(my_data)

## 'data.frame':    60 obs. of  3 variables:
##  $ len : num  4.2 11.5 7.3 5.8 6.4 10 11.2 11.2 5.2 7 ...
##  $ supp: Factor w/ 2 levels "OJ","VC": 2 2 2 2 2 2 2 2 2 ...
##  $ dose: num  0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 ...
```

From the output above, R considers “dose” as a numeric variable. We'll convert it as a factor variable as follows:

```
my_data$dose <- factor(my_data$dose,
                      levels = c(0.5, 1, 2),
                      labels = c("D0.5", "D1", "D2"))
head(my_data)
```

```
##      len supp dose
## 1  4.2   VC D0.5
## 2 11.5   VC D0.5
## 3  7.3   VC D0.5
## 4  5.8   VC D0.5
## 5  6.4   VC D0.5
## 6 10.0   VC D0.5
```

A look at the frequency table shows that we have 2x3 design cells with the factors being supp and dose and 10 subjects in each cell. Here, we have a *balanced design*.

```
table(my_data$supp, my_data$dose)
```

```
##
##      D0.5 D1 D2
##  OJ    10 10 10
##  VC    10 10 10
```

Compute summary statistics

Next we will compute mean and SD by groups using dplyr R package:

```
require("dplyr")
group_by(my_data, supp, dose) %>%
  summarise(
    count = n(),
    mean = mean(len, na.rm = TRUE),
    sd = sd(len, na.rm = TRUE)
  )
```

```
## # A tibble: 6 x 5
## # Groups:   supp [2]
##   supp dose count mean  sd
##   <fct> <fct> <int> <dbl> <dbl>
## 1 OJ   D0.5    10 13.2  4.46
## 2 OJ   D1      10 22.7  3.91
## 3 OJ   D2      10 26.1  2.66
## 4 VC   D0.5    10  7.98  2.75
## 5 VC   D1      10 16.8  2.52
## 6 VC   D2      10 26.1  4.80
```

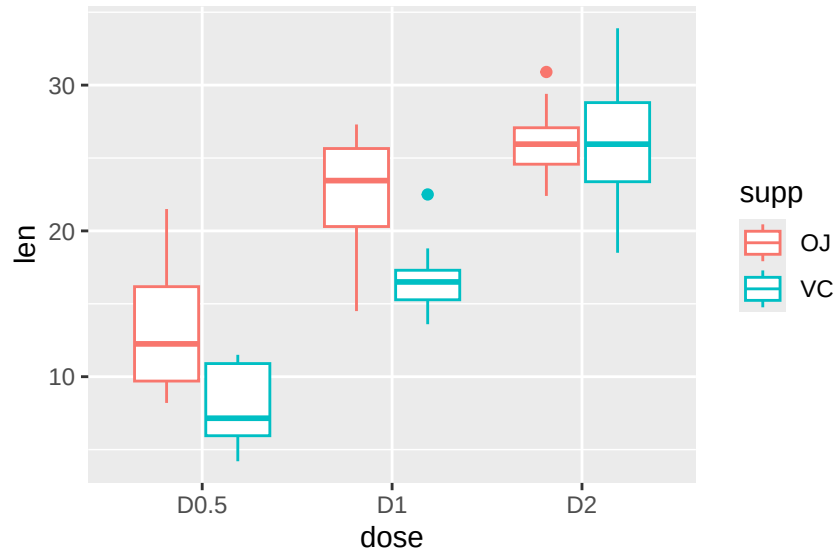
Visualise the data

Box plots and line plots can be used to visualize group differences:

- **Box plot** to plot the data grouped by the combinations of the levels of the two factors.
- **Two-way interaction plot**, which plots the mean (or other summary) of the response for two-way combinations of factors, thereby illustrating possible interactions.

To visualise our data with ggplot2:

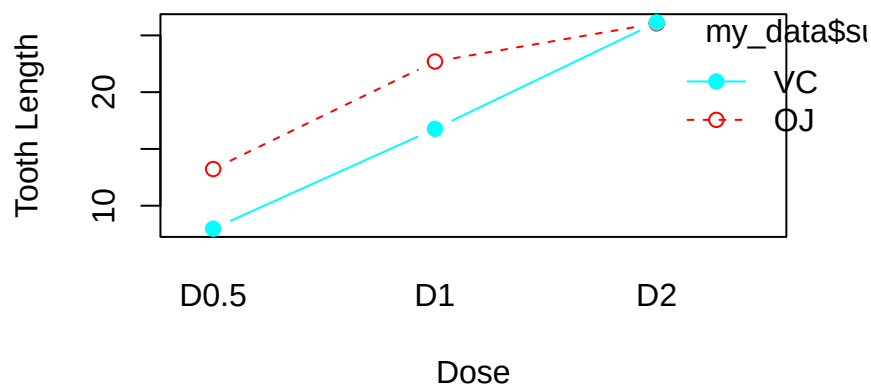
```
library(ggplot2)
ggplot(my_data, aes(x=dose, y=len, color=supp)) + geom_boxplot()
```



The nice thing about the above command is that allows to separately plot boxes for the two types of supplement. An alternative, and more commonly used, visualisation for data with two factors is to use an “interaction plot”.

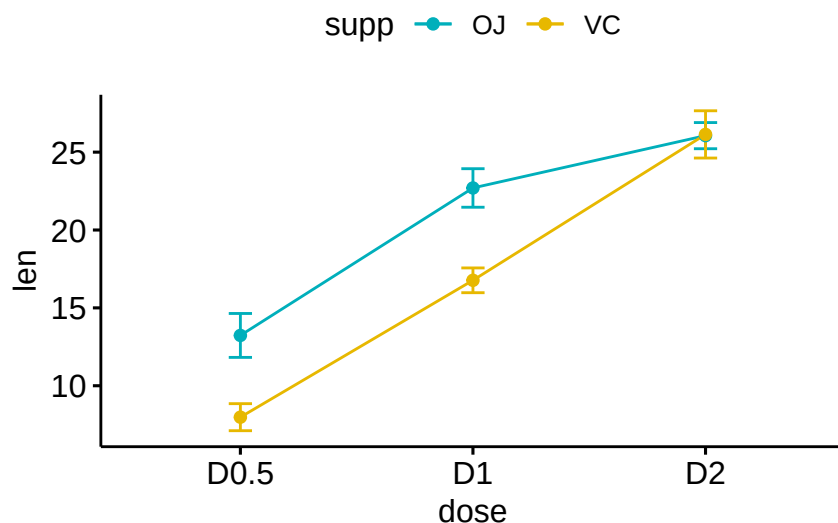
The basic interaction plot command is `interaction.plot()`.

```
interaction.plot(x.factor = my_data$dose, trace.factor = my_data$supp,
  response = my_data$len, fun = mean,
  type = "b", legend = TRUE,
  xlab = "Dose", ylab="Tooth Length",
  pch=c(1,19), col = rainbow(2))
```



More advanced interaction plots that include error bars can be created using the “ggpubr” package.

```
library("ggpubr")
ggline(my_data, x = "dose", y = "len", color = "supp",
  add = c("mean_se"),
  palette = c("#00AFBB", "#E7B800"))
```



Note that “mean_se” will plot the mean and the standard error. We can also plot other error bars such as standard deviation (“mean_sd”) or confidence interval (“mean_ci”).

Computing the two-way ANOVA

We want to know if tooth length depends on supp and dose. To run the two way, between-subject ANOVA, we will use the `aov()` function. The basic syntax is `aov(formula, data = NULL, projections = FALSE, qr = TRUE, contrasts = NULL, ...)`. The way the formula is expressed is “<dependent variable> ~ <factor 1> * <factor 2>”.

```
res.aov2 <- aov(len ~ supp * dose, data = my_data)
```

As before, the function `summary.aov()` is used to summarize the analysis of variance model.

```
summary(res.aov2)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## supp       1   205.4    205.4  15.572 0.000231 ***
## dose       2  2426.4   1213.2   92.000 < 2e-16 ***
## supp:dose   2   108.3     54.2    4.107 0.021860 *
## Residuals  54   712.1     13.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The output includes the columns *F value* and *Pr(>F)* corresponding to the p-value of the test.

From the ANOVA results, you can conclude the following, based on the p-values and a significance level of 0.05: * the p-value of supp is 0.000231 (significant), which indicates that the levels of supp are associated with significant different tooth length. * the p-value of dose is < 2e-16 (significant), which indicates that the levels of dose are associated with significant different tooth length. * the p-value for the interaction between supp*dose is 0.02 (significant), which indicates that the relationships between dose and tooth length depends on the supp method.

In summary, it can be seen that we have two significant main effects, but also a significant *interaction* effect. The significant interaction effect means, that we *cannot* make any statement about the effect of dose of supplement!

Multiple pairwise-comparison between the means of groups

In ANOVA test, a significant p-value indicates that some of the group means are different, but we don't know which pairs of groups are different. It's possible to perform multiple pairwise-comparison, to determine if the mean difference between specific pairs of group are statistically significant. To identify where the significant differences are, we can use the same methods as for the one-way ANOVA.

We do not need to perform the test for the "supp" variable because it has only two levels, which have been already proven to be significantly different by ANOVA test. Therefore, the Tukey HSD test will be done only for the factor variable "dose".

```
TukeyHSD(res.aov2, which = "dose")
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = len ~ supp * dose, data = my_data)
##
## $dose
##          diff          lwr          upr    p adj
## D1-D0.5  9.130   6.362488 11.897512 0.0e+00
## D2-D0.5 15.495  12.727488 18.262512 0.0e+00
## D2-D1    6.365   3.597488  9.132512 2.7e-06
```

How to read the table:

- **diff**: difference between means of the two groups
- **lwr**, **upr**: the lower and the upper end point of the confidence interval at 95% (default)
- **p adj**: p-value after adjustment for the multiple comparisons.

It can be seen from the output, that all pairwise comparisons are significant with an adjusted p-value < 0.05.

Pairwise t-test

As before, we can also use the function `pairwise.t.test()` to calculate pairwise comparisons between group levels with corrections for multiple testing.

```
pairwise.t.test(my_data$len, my_data$dose,
                p.adjust.method = "BH")
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data: my_data$len and my_data$dose
##
##      D0.5      D1
## D1 1.0e-08 -
## D2 4.4e-16 1.4e-05
##
## P value adjustment method: BH
```

Checking ANOVA assumptions

The assumptions for the two-way ANOVA are exactly the same as for the one-way ANOVA: Homogeneity of variance and normality.

Homogeneity of variance

```
library(car)
leveneTest(len ~ supp*dose, data = my_data)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  5  1.7086 0.1484
##      54
```

From the output above we can see that the p-value is not less than the significance level of 0.05. This means that there is no evidence to suggest that the variance across groups is statistically significantly different. Therefore, we can assume the homogeneity of variances in the different treatment groups.

Normality

```
# Extract the residuals
aov_residuals <- residuals(object = res.aov2)
# Run Shapiro-Wilk test
shapiro.test(x = aov_residuals )

##
##  Shapiro-Wilk normality test
##
## data:  aov_residuals
## W = 0.98499, p-value = 0.6694
```

The *Shapiro-Wilk test* on the ANOVA residuals ($W = 0.98, p = 0.5$) finds no indication that normality is violated.

Exercise 3

For this exercise, we will use import the data from the file “data/twowayanovalong.csv”, the convert “year” and “genotype” to factors:

```
my_data <- read.csv("Lab04_data.csv")
my_data$year <- as.factor(my_data$year)
my_data$genotype <- as.factor(my_data$genotype)
```

Your tasks:

1. Plot the effect of the factors “year” and “genotype” on “value” using the `ggline()` function
2. Show the frequency table for “year” and “genotype” using the `table()` function
3. Calculate a two-way ANOVA with factors “year” and “genotype”
4. Perform a Tukey HSD post-hoc test on the significant factors

Further reading

- [Two-Way ANOVA Test in R](#)
- [MANOVA Test in R: Multivariate Analysis of Variance](#)
- [Kruskal-Wallis Test in R \(non parametric alternative to one-way ANOVA\)](#)
- [Quick-R: ANOVA/MANOVA\]](#)
- [Quick-R: \(M\)ANOVA Assumptions\]](#)
- [R and Analysis of Variance](#)